

NDATrace: When Less Architecture Produced the Stronger Result

An evidence-grounded NDA review study of semantic reasoning, retrieval boundaries, rejected agentic complexity and the economics of mandatory human verification.

Ubaidulla Asmitha | 2 October 2026 | Frozen experiments; no new model inference for this report

Word counts: analytical narrative 1,114 · full Sections 1-8 3,369 · complete document 3,789

Length disclosure. The analytical narrative is within the 1,200-word ±15% tolerance. The table-inclusive main report is substantially longer because this consolidated version retains the required experiment matrix, decision tables and captions; it should not be represented as a 1,200-word complete document.

1. The business problem and actual contribution

Tina, an enterprise legal-operations analyst, currently reads a vendor NDA against 17 confidentiality requirements before counsel decides what to do. Keyword search is a weak substitute: the official rule baseline found only 16.8% of contradictions because obligations can be paraphrased, qualified by exceptions or assembled across clauses. NDATrace changes the hand-off, not the legal authority. It proposes a structured label, exposes the quoted clauses and lets Tina record an approval, override or rejection. That demonstrated workflow justifies semantic AI over rules while keeping verification human. I measured classification, evidence and operating signals; I did not measure time saved, legal outcomes or realised productivity.

2. Architecture and build-versus-rent decisions



Figure 1. Current frozen path, verified against the runtime modules. Blue stages retrieve, orange rents semantic inference, green stages enforce deterministic controls, and the reviewer remains authoritative.

TABLE 1. OWNERSHIP ACROSS THE IMPLEMENTED STACK

Layer	Decision	Implemented choice	Boundary or gap
Compute	Rent / local comparison	OpenRouter inference; Ollama for local experiments	Provider dependency; local energy/compute not monetised
Model	Rent	GPT-5-mini, temperature 0	No fine-tuning; behaviour remains probabilistic
Data	Reuse	ContractNLI, CC BY 4.0	Public, short NDAs are not enterprise validation
Retrieval	Build + reuse libraries	Clause chunks, BM25, cross-encoder	Project owns configuration, not pretrained reranker
Orchestration	Build	Single bounded call; no agent or confidence gate	No automatic FULL fallback
Evaluation	Build	Split discipline, paired tests, Joint evaluator	Gold overlap is an imperfect legal proxy
Serving / interface	Build	FastAPI + Next.js + persisted decisions	Replaced the proposed thin Streamlit demo to support review history and a real decision API
Governance / observability	Build, incomplete	Source checks, injection flags, rate/concurrency/cost limits, logs	No enterprise identity, access policy or production monitoring

I placed semantic interpretation inside one rented model call and kept provenance, parsing, resource limits and final authority outside it. This makes failures observable and prevents the model from validating its own claims. The cost is integration complexity and dependence on a hosted model whose reasoning I cannot fully control. Retrieval is also an ownership choice: bounded context is predictable to serve, yet omitting a decisive clause becomes my system's failure. Reusing mature ranking libraries reduced implementation risk, but their pretrained objectives may not reflect legal relevance. A FastAPI/Next.js split replaced the proposed thin Streamlit interface because recorded reviewer decisions and history required a persistent backend rather than a disposable demonstration shell.

3. The complete experimental journey

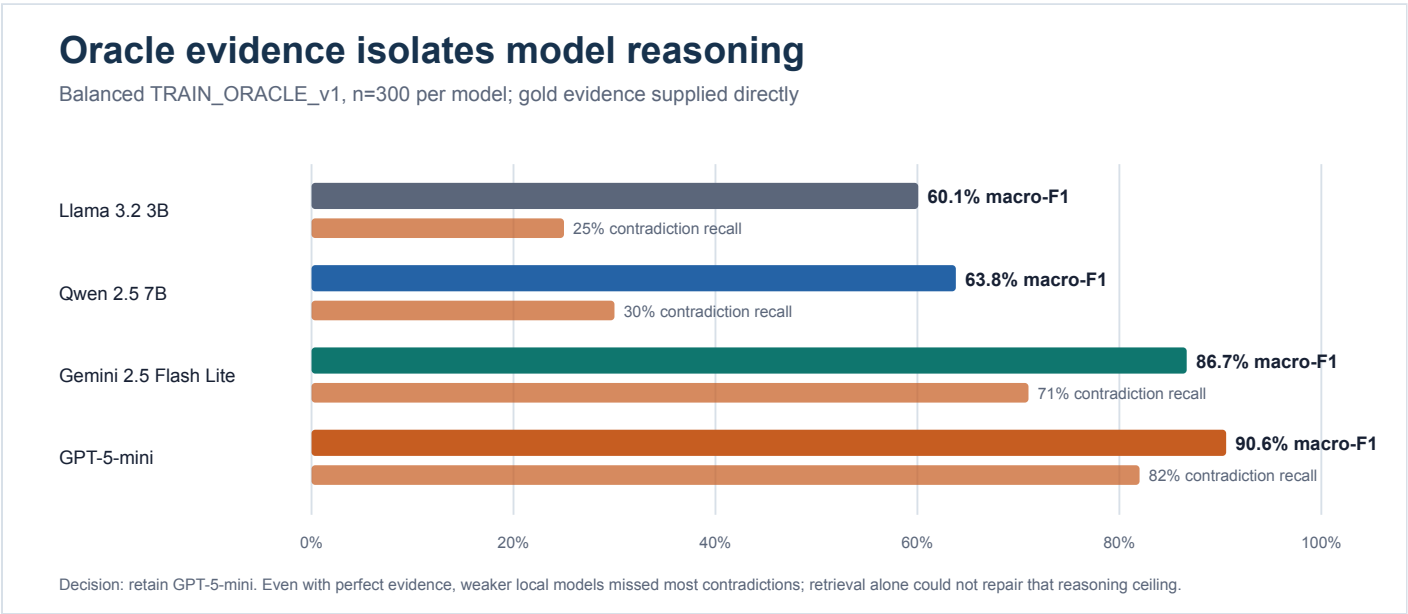


Figure 2. Oracle evidence removes retrieval as the explanation for error. GPT-5-mini's stronger macro-F1 and contradiction recall justified renting it despite higher latency and cost; the Oracle score is a diagnostic ceiling, not a deployable result.

TABLE 2. MODEL AND PROMPT SELECTION

Engineering question	Alternatives and population	Critical evidence	Decision and accepted limitation
Which reasoner should receive retrieval effort?	Llama, Qwen, Gemini and GPT with gold evidence; balanced TRAIN n=300/model	GPT macro-F1 0.906, contradiction recall 82%; Qwen 0.638 and 30%	GPT-5-mini. Hosted dependency and higher output cost accepted for the measured reasoning advantage.
How much instruction helps?	Minimal P0, definitions P1 and procedure P2; Qwen TRAIN n=150; later GPT controls	P0 led Qwen macro-F1 0.507 and contradiction recall 22%; added structure increased C→NM errors; GPT confirmation did not establish a replacement	Freeze P0. Prompt complexity must show repeatable gain, not plausibility.



Figure 3. Retrieval decisions are separated by stage. Fixed-512’s apparent recall win doubled mean context to about 1,986 tokens and depressed MRR; after a common reranker, BM25 and dense retrieval were effectively tied. Simplicity, not claimed superiority, selected BM25.

TABLE 3. RETRIEVAL CONFIGURATION DECISION

Component	Alternatives	Measured implication	Frozen choice
Chunk boundary	Sentence, clause-256, fixed-512	64.2%, 87.0%, 97.8% Recall@5 respectively; fixed-512 saturated by returning near-whole documents and MRR fell to 0.183	Clause-aware 256 tokens, no effective overlap
Candidate generator	BM25, dense BGE, hybrid RRF	Pre-rerank Recall@5 90.5%, 88.4%, 92.1%; after identical reranking BM25 and dense differed on 1/4,371 cases	BM25 top-20; avoid an unearned vector index
Final context	K=3, 5, 10; top-11 classification test	MRR flattened after K=5; top-11 Joint +4 points but p=0.263 and input +64%	Cross-encoder top-5; wider context rejected

TABLE 4. COMPREHENSIVE EXPERIMENTAL DECISION MATRIX

Experiment and population	Decisive experimental evidence	Engineering decision
STAGE 1 ESTABLISH INTELLIGENCE		
Majority and deterministic Rule baselines Official TEST · n=2,091 E04B · final comparison	Majority: 46.3% accuracy; 0% Joint. Rule: 59.0% accuracy; 50.1% Joint; 16.8% Contradiction recall.	Retain Rule as a comparator, not the product. Semantic interpretation is required, but the zero-API-cost floor remains visible.
Oracle model comparison Balanced TRAIN diagnostic · n=300/model · gold evidence E01 notebook · metrics	GPT-5-mini: Macro-F1 0.906; Contradiction recall 82%. Gemini: 0.867; 71%. Qwen: 0.638; 30%.	Select hosted GPT-5-mini. Accept provider dependency and higher cost for the strongest measured reasoning ceiling; gold evidence still leaves reasoning error.

Experiment and population	Decisive experimental evidence	Engineering decision
Controlled prompt variants Qwen TRAIN n=150; GPT TRAIN n=150 + fresh confirmation n=150 E03 · E12B · E12C	Qwen P0/P1/P2: Macro-F1 0.507/0.448/0.403; Contradiction recall 22%/6%/2%. GPT P3 versus P0: +8 Joint cases initially, -3 on document-disjoint confirmation; pooled +1.7 points, 95% CI [-1.7, +5.0].	Retain minimal P0. Added instruction complexity did not produce a repeatable gain.
STAGE 2 OPTIMIZE EVIDENCE		
Chunk-boundary experiment Evidence-bearing TRAIN · n=4,371 E06 notebook · summary	Recall@5 / MRR: clause-256 87.0%/0.276; sentence 64.2%/0.441. Fixed-512: 97.8% recall but MRR 0.183 because near-whole documents saturated top-5.	Choose clause-aware 256-token chunks. Preserve useful clause units without treating document saturation as retrieval quality; configured overlap 50 is not applied by the clause-aware chunker.
Candidate retrieval and cross-encoder reranking Matched TRAIN · n=4,371 · common clause-256 candidates E06 summary · matched control	Before reranking: BM25/dense/hybrid Recall@5 90.5%/88.4%/92.1%. After top-20 reranking: all ≈92.2%; BM25 and dense differed on 1/4,371 cases. Reranking recovered 231 and regressed 72 versus dense alone.	Select BM25 top-20 → cross-encoder top-5. Reranking earns its cost; a dense index and hybrid fusion do not after reranking.
Top-K static context expansion Matched TRAIN · top-5 versus top-11 · n=150 E12A notebook · analysis	Top-11: Joint +4.0 points, but p=0.263; input tokens +64%. Recoveries were offset by distractor-driven regressions.	Keep top-5 context. Wider static context increased cost without establishing a reliable quality gain.
STAGE 3 SELECT ARCHITECTURE		
FULL versus RAG Official TEST · matched n=2,091 · same GPT prompt/evaluator E20 report · summary	Joint: FULL 74.6% versus RAG 72.5%; paired McNemar p=0.0047. RAG efficiency: input tokens -50.4%; inference cost -16.8%.	Keep FULL as the measured quality reference; retain RAG as the bounded-context prototype. This accepts a measured Joint disadvantage for predictable context and lower inference cost.
STAGE 4 TEST ADDED AUTONOMY		
Selective-agent evaluation TRAIN n=150 · 15 routed; 135 frozen base outputs E11 notebook · paired comparison	Joint: 74.0% → 74.0%; one recovery and one regression; p=1.0. Behaviour: 0 tool calls across 15 triggers; \$0.0218 incremental spend.	Reject the selective agent. Routing incurred cost but produced zero net Joint value under this historical evaluator protocol.
Forced full-agent and investigation-prompt tests Same TRAIN n=150 · every case enters agent · current evaluator v2 full-agent · prompt V2	Full-agent V1: Joint 75.3% → 73.3%; tools used on 2/150; 0 useful tool recoveries. Investigation V2: tools used on 30/150, but useful tool-associated recoveries remained 0; Joint fell to 68.7%.	Reject the tested agent, not merely its router. Forcing investigation changed tool-use behaviour but reduced evidence-grounded quality.
Automatic confidence/review routing Fresh DEV validation · n=138 · R1-R3 policies E15 notebook · results	R3: reviewed 51.4%, captured 82.5% of Joint failures, left 10.4% residual Joint error. It missed both provisional criteria: ≤40% review and <10% residual error.	Reject automatic confidence routing. Retain R1 only for structural/source-integrity flags and require human verification.
STAGE 5 VALIDATE OPERATIONS		
Adversarial robustness and safeguards Curated diagnostic · 20 clean/attack pairs; 11 injection pairs E16 · E22 controls	Hosted diagnostic: Joint 85% clean → 75% attack; 4/11 injection patterns succeeded. Controls: resource limits passed targeted regression; the pattern-based injection guard remained partial.	Quarantine suspicious inputs and require human verification. Source-valid quotations do not prove that embedded instructions are trustworthy.
Targeted difficult-case evaluation Curated DEV regression battery · n=49 · not representative E24 notebook · analysis	Joint: Rule 44.9%; FULL 73.5%; RAG 67.3%. Three of four documented exception/carve-out cases still failed; multi-span evidence losses persisted.	Maintain a mechanism-specific regression suite. Do not generalise this deliberately hard DEV battery to the official TEST population.
Token economics and cost-to-serve model Measured official TEST costs + explicitly hypothetical volume/fallback scenarios E20 measured inputs · E18 model	Measured per requirement: FULL 2,279 input tokens/\$0.00202; RAG 1,131/\$0.00168; output ≈727/701 tokens. Counterfactual at 500 NDAs/year: selective human fallback yields FULL \$7,212 versus RAG \$7,819 under the declared 17-requirement, 5-minute, \$40/hour assumptions.	Optimize evidence quality before inference cents, while requiring human verification in the actual workflow. Keep measured operating cost separate from hypothetical business scenarios.

I tested the reasoning ceiling first because optimising retrieval around an inadequate reasoner would misdiagnose the constraint. I then froze model, prompt and retrieval choices before comparing architectures, so each later difference had a narrower explanation. That discipline was difficult: early runs used different models, balanced samples and historically exposed DEV/TEST cases, while a superseded lineage had partially observed TEST. I therefore use earlier work only for mechanism and reserve the re-derived 2,091-case run for final comparisons. I also corrected the evaluator so full-

document access was never counted as evidence by construction; the system had to return an explicit overlapping quotation. Negative results were productive: they removed dense retrieval, prompt elaboration, a wider context, the agent and confidence routing rather than letting plausible complexity accumulate.

4. The unexpected architecture result and metric critique

TABLE 5. UNIFIED ARCHITECTURE COMPARISON

Population / metric	Rule	FULL	RAG
Official TEST - n=2,091			
Accuracy	59.0%	77.6%	76.8%
Macro-F1	0.479	0.727	0.723
Joint correctness	50.1%	74.6%	72.5%
Contradiction recall	16.8%	75.5%	77.3%
NotMentioned recall	90.5%	62.7%	63.2%
Risk-sensitive recall (C/NM mean)	53.6%	69.1%	70.3%
Mean input tokens / API cost	n/a / \$0 API	2,279 / \$0.00202	1,131 / \$0.00168
Mean latency	0.06 ms local	7.31 s	7.14 s
Targeted DEV - n=49 (diagnostic, not pooled)			
Accuracy / Macro-F1	51.0% / 0.457	73.5% / 0.720	71.4% / 0.708
Joint / Contradiction recall	44.9% / 13.3%	73.5% / 46.7%	67.3% / 53.3%

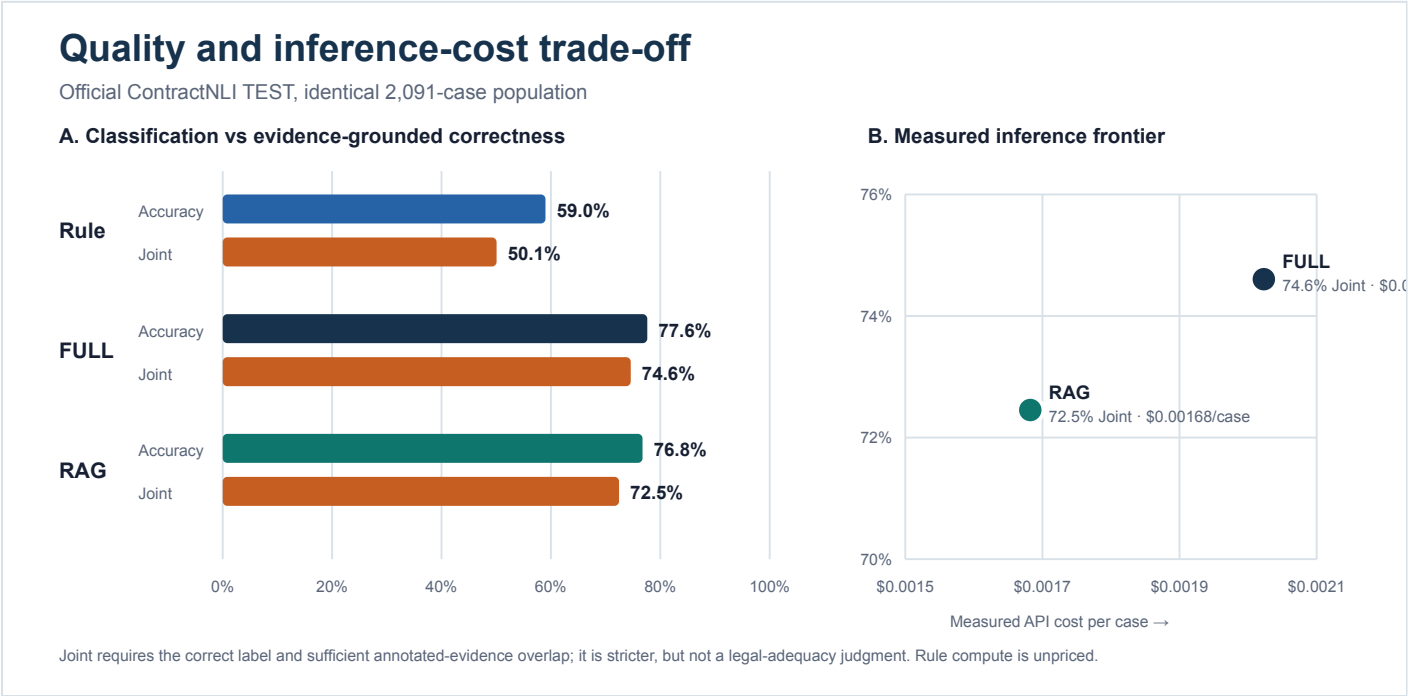


Figure 4. Left: accuracy overstates evidence-grounded performance for every architecture. Right: the quality-cost frontier shows why RAG is an operating trade-off rather than the quality winner. Both panels use the common official TEST population; Rule compute cost is unpriced.

I expected selective retrieval to improve efficiency and possibly evidence quality by removing irrelevant clauses. The first expectation held; the second did not. FULL's 2.1-point Joint advantage was statistically significant even though its accuracy advantage was not. I retained RAG because its context is bounded, its measured inference cost is lower and

case 038 shows that excluding a permissive distractor can help. The counter-evidence matters equally: cases 007 and 043 lost or failed to cite necessary spans. ContractNLI agreements are relatively short, so better scaling on long enterprise contracts remains a deployment hypothesis, not a result.

Accuracy cannot establish that Tina received defensible evidence. Joint correctness adds that constraint, but annotated-span overlap still cannot judge whether an explanation is legally adequate, whether the benchmark annotation is contestable or whether a reviewer would accept an alternative clause. Its threshold also treats partial multi-span coverage as a binary outcome. Class-specific recall is necessary because a missed contradiction and an unnecessary escalation have different consequences; averaging them can conceal the rarer class. Finally, public benchmark labels encode fixed hypotheses and annotation choices rather than an organisation's policy, jurisdiction or risk tolerance. The evaluation supports comparative engineering decisions; it does not validate autonomous legal judgment.

5. Agent experiments, failure investigation and rejected complexity

TABLE 6. AGENT PROGRESSION - MATCHED TRAIN EVIDENCE, NEVER POOLED WITH TEST

Evaluation	Exposure and tool use	Outcome vs. matched control	Interpretation
Selective V1	15/150 routed; 0 tool calls	Joint 74.0% → 74.0% (1 recovery, 1 regression, net 0); +\$0.0218, +6.07 s/escalation	Controller suppressed investigation
Full-agent V1	All 150 forced through agent; 2/150 tool cases, 0/2 useful	Joint 75.3% → 73.3% (net -3 cases); +\$0.001515/case, +5.87 s/case	Removing routing did not create value
Investigation-prompt V2	Same 150, prompt encouraged tool use; 30/150 tool cases, 39 calls, 0/30 useful	Joint 75.3% → 68.7% (net -10 cases); +\$0.002659/case, +10.36 s/case, 25 fallbacks	Tool use rose; outcomes worsened

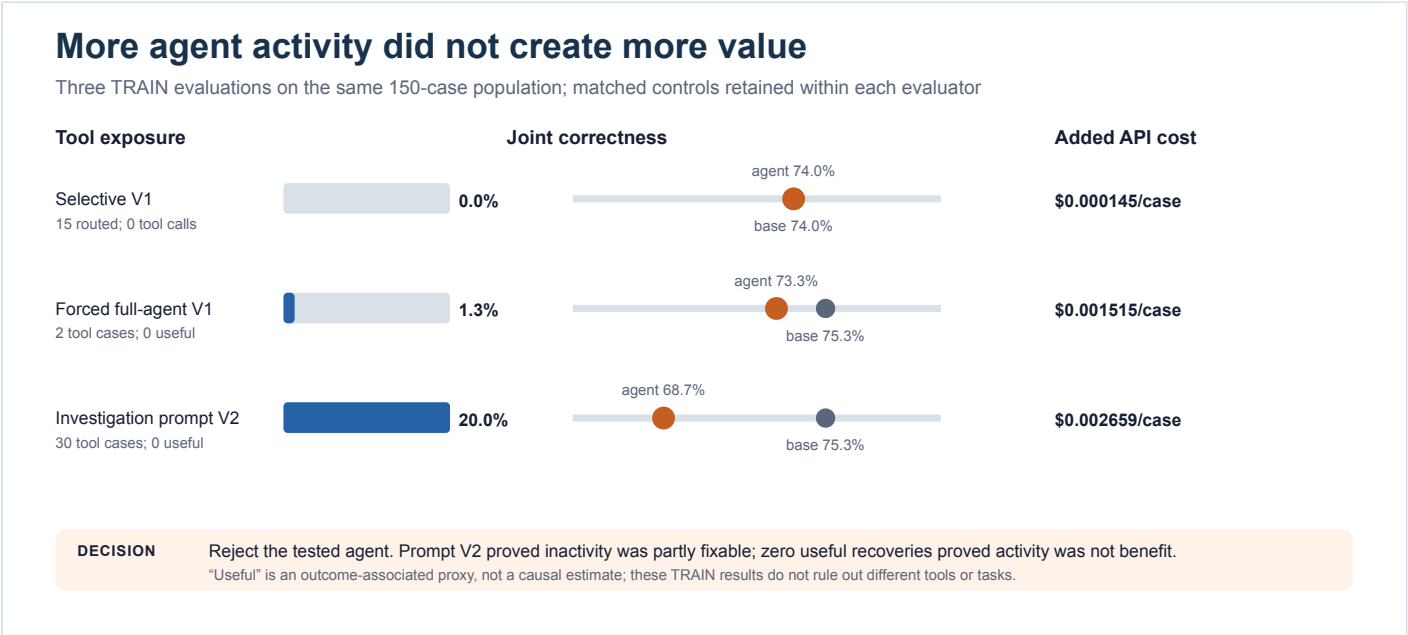


Figure 5. Increasing investigation changed agent behaviour but not its value. The selective trial used its own matched evaluator; the forced V1/V2 comparison uses the later evaluator-v2 control. Results are never pooled across evaluators.

TABLE 7. ROUTING WORKLOAD VERSUS RESIDUAL JOINT ERROR - VALIDATION DEV N=138

Policy signal	Review rate	Failure capture	Residual error	Interpretation
R1 structural/source validity	4.3%	12.5%	26.5%	Useful guard, poor uncertainty detector
R2 label/rule signal	21.0%	37.5%	22.9%	Missed all four contradiction failures
R3 combined	51.4%	82.5%	10.4%	Missed both provisional ≤40% workload and <10% error criteria

Review workload did not buy a safe automation region

E15 validation DEV n=138; provisional target: review ≤40% and residual Joint error <10%



Figure 6. No tested policy enters the jointly specified workload/error region. R3 captured more failures, but only by reviewing over half the cases and still missing the residual-error criterion.

RAG residual failures are mostly interpretive

Official TEST: 576 mutually exclusive non-Joint cases

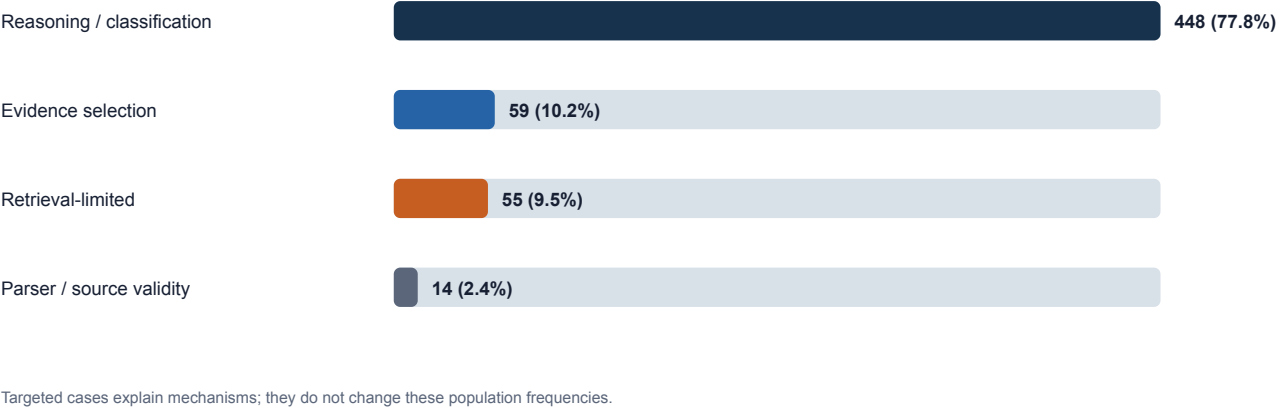


Figure 7. Population-level RAG failures on official TEST. Categories sum to 576 and are not mixed with curated case diagnoses.

TABLE 8. TARGETED MECHANISMS - DIAGNOSTIC DEV EVIDENCE ONLY

Mechanism	Verified cases	What the contrast revealed
Partial retrieval + incomplete citation	007, 043	RAG omitted some gold spans and failed to cite some available spans
Exception / override reasoning	034, 039, 040	FULL and RAG shared the same interpretive failure
Distractor avoidance	038	RAG excluded a permissive clause that misled FULL
Annotation caveat	001	The benchmark answer itself is substantively debatable

The first selective controller reached FINAL on all 15 triggers, so I tested the alternative explanation that routing or prompt reluctance—not agency itself—caused failure. Forcing all 150 cases through V1 produced only two tool-using cases and no useful recovery. A V2 prompt then raised tool use to 20%, but useful tool-associated recovery remained zero, Joint fell

by 6.7 points, and fallbacks increased. The effectiveness label is outcome-associated, not causal, and all three evaluations are small TRAIN studies. They nevertheless reject the tested agent for lack of value rather than mere tool inactivity; different tools or challenge sets remain untested.

Routing failed a related test. Structural errors were detectable, but semantic mistakes often looked ordinary: the strongest policy still required half the cases to be reviewed and narrowly missed its residual-error threshold. Confidence was not calibrated, and Rule agreement could not expose cases where both systems were wrong alike. The 138-case validation also left wide uncertainty around the residual estimate, so the workload failure was the clearer rejection reason. I therefore made human verification mandatory rather than presenting weak routing as safe automation.

6. Cost-to-serve and business reasoning

TABLE 9. MEASURED ARCHITECTURE ECONOMICS - OFFICIAL TEST N=2,091

Measure	FULL	RAG
Mean input tokens	2,279	1,131
Mean output tokens	~727	701
Measured API cost / case	\$0.00202	\$0.00168
Mean latency	7.31 s	7.14 s
Joint correctness	74.6%	72.5%

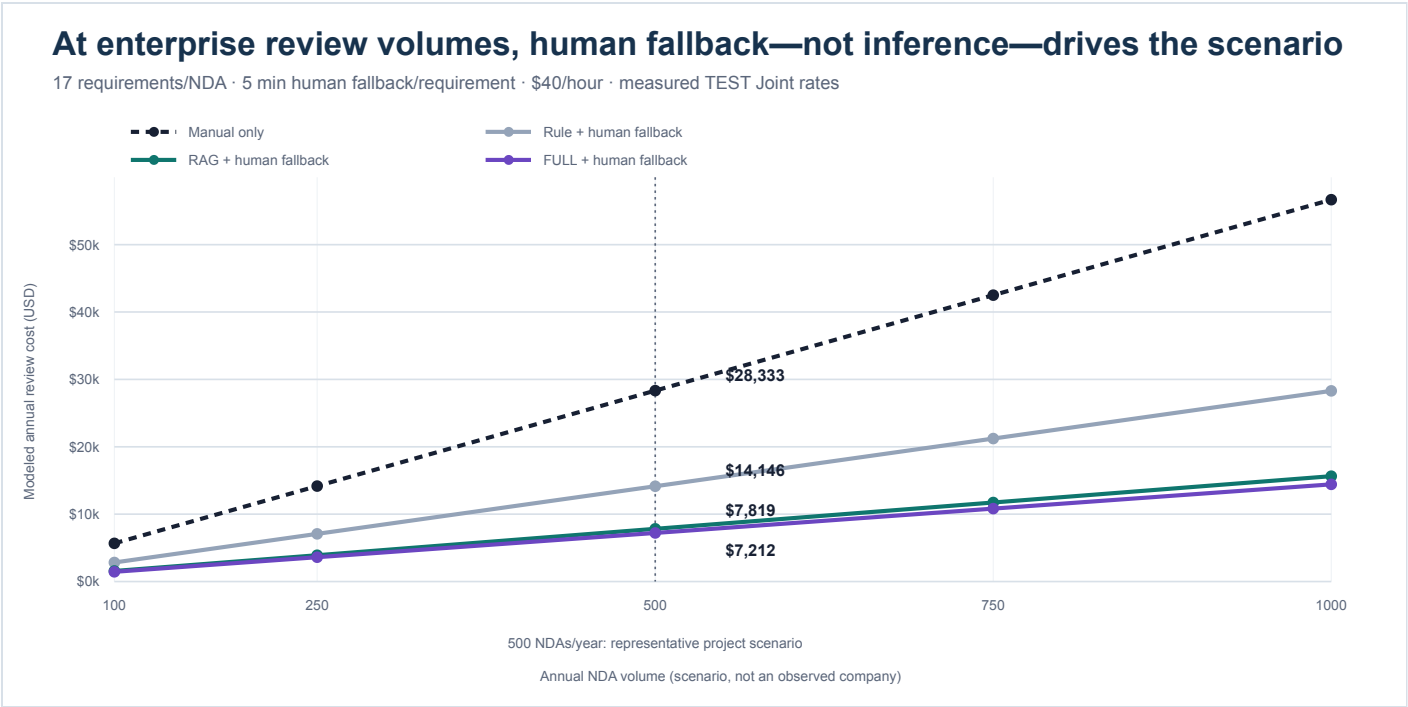


Figure 8. Counterfactual selective-fallback scenario: 17 requirements per NDA, five minutes of human fallback per requirement at \$40/hour, and measured official-TEST Joint rates. At 500 NDAs/year the modeled totals are \$28,333 manual, \$14,146 Rule, \$7,819 RAG and \$7,212 FULL. These are formula outputs—not observed company expenditure—and assume Joint-success cases avoid fallback, unlike NDATrace's actual mandatory-review policy.

Interpretation boundary. This scenario answers “what if human review occurred only after a non-Joint result?” It illustrates why human-time assumptions dominate inference cost; it does not establish savings, productivity or the deployed review workflow.

TABLE 10. DECLARED COST-MODEL ASSUMPTIONS

Component	Base input	Status and accounting rule
Workload	1,000 requirement reviews	Scenario volume
Mandatory verification	5 min/case at \$40/hour	Applied once to every architecture; not avoided by correctness
Incremental investigation	10 min × non-Joint rate	Hypothetical proxy after base verification; not measured and not overlapping
Inference	Saved TEST cost/case	Measured API spend; Rule local compute remains unpriced
Fixed operations	Excluded	Hosting, monitoring and support were not measured; exclusion is not zero cost

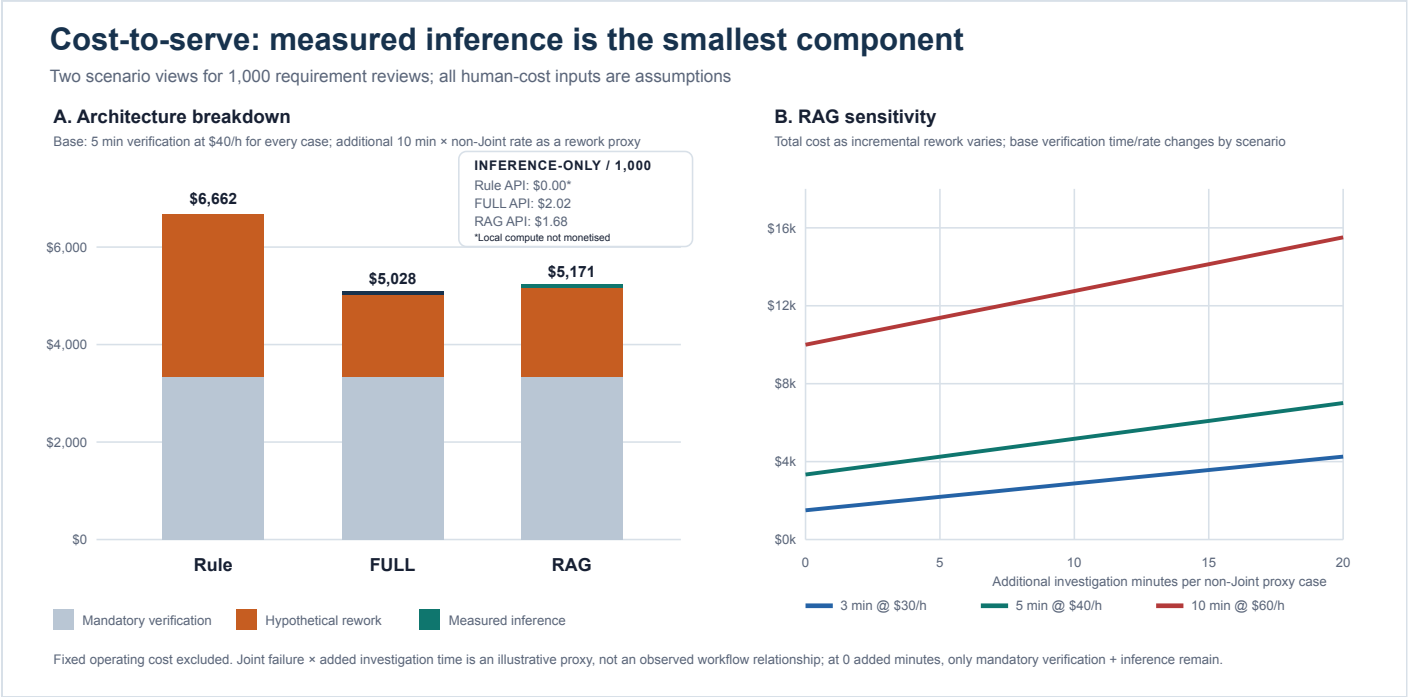


Figure 9. Formula: $V \times [\text{measured inference} + \text{mandatory verification} + (1 - \text{Joint}) \times \text{assumed additional investigation}] + \text{fixed}$. Base panel: $V=1,000$, five verification minutes at \$40/hour for every system and ten additional minutes per non-Joint proxy case; fixed cost excluded. Sensitivity varies verification time/rate and added investigation from zero to twenty minutes. Joint failure is an assumption for investigation, not an observed relationship.

RAG halved input tokens, yet output stayed near 700 tokens per case; the professor’s warning was therefore borne out: generated evidence and explanation materially limit the saving from shorter context. Latency barely changed, which further weakens a simple “fewer tokens means faster” claim. Once labour is introduced, the sub-cent inference difference becomes almost irrelevant. The plotted ranking changes when Joint failure is assumed to trigger extra investigation; at zero added investigation, architecture differences reduce almost entirely to measured inference. Neither view proves the cheaper enterprise workflow because review time, error discovery and follow-up are unmeasured. Fixed hosting, monitoring and support are excluded rather than invented. A timed reviewer crossover must record verification time, overrides, escalations, accepted evidence and follow-up investigation before any operating-cost or ROI conclusion.

7. Critical evaluation, difficulties and future development

TABLE 11. LIMITATIONS TRANSLATED INTO VALIDATION WORK

Limitation	What experiments established	Current mitigation	Next validation step
Exception and polarity reasoning	Dominant reasoning failures persist across FULL and RAG	Visible evidence; mandatory reviewer	Research 1: independent exception/override battery
Multi-span evidence	Retrieval and citation can each lose necessary spans	Top-20 reranking; Joint scoring	Research 2: separate coverage and citation-recall evaluation
Short public benchmark	RAG cost is bounded; long-document advantage untested	Claim restricted to ContractNLI	Research 3: longer, varied agreements with expert labels
Unknown workflow impact	No reviewer-time or override study exists	Economics labelled scenario-only	Business 1: timed crossover reviewer study
Prompt injection	4/11 patterns succeeded; source-valid text can still be hostile	Pattern flag, quarantine, limits	Research 4: broader adaptive attack evaluation
Enterprise controls	Prototype logs decisions but is not production-governed	Public data only	Production: identity, access, retention and monitoring

The hardest engineering task was separating evidence access from interpretation. Oracle evidence, retrieval diagnostics and case-level traces let me distinguish missing clauses from wrong reasoning, but Joint still compresses several mechanisms into one score. Controlled comparison was another difficulty: changing model, prompt, context and population together would have made the architecture verdict uninterpretable, so I froze layers and disclosed historical exposure rather than overstating evaluation blindness. Budget controls forced forecasts, compact outputs and bounded runs, while latency made every extra call operationally visible.

Important problems remain unresolved. Exception clauses still defeat the strongest architecture; the agent's non-use of tools explains its rejection but not how to build a useful investigator; routing signals do not isolate semantic risk; and a source-valid quotation can contain adversarial instructions. Resource limits contained cost and denial-of-service exposure, but they did not make document instructions trustworthy. My priorities therefore separate research from production: improve exception reasoning and multi-span evidence first, validate on longer agreements and with reviewers next, then expand security testing. Identity, confidentiality controls and monitoring are prerequisites for deployment, not substitutes for better evidence.

8. Conclusion

NDATrace established that semantic AI adds substantial value beyond keyword rules, but additional architecture did not reliably add value beyond the core model-and-evidence path. FULL remained the strongest quality reference; RAG was retained only as a bounded-context operating trade-off, while the agent and automatic routing were removed. The result is a working evidence-grounded reviewer-assistance prototype with explicit human authority. The consequential open question is whether RAG earns its evidence losses on longer, more varied enterprise agreements.

Supporting experimental evidence and reproducibility

The analytical narrative above is intended to stand alone. Detailed experiment artifacts remain linked from the comprehensive matrix rather than reproduced wholesale. The source-of-truth order is: submitted objectives for intended targets; saved canonical results for measurements; executed notebooks and decision records for context; project UI for presentation cues only.

Population discipline. Official TEST n=2,091 supplies the headline Rule/FULL/RAG comparison. TRAIN samples support model, prompt, retrieval, context and agent decisions. DEV n=138 supports routing validation; curated DEV n=49 diagnoses mechanisms. These populations are labelled and never pooled.

Figure selection. The nine regenerated SVGs answer distinct questions: implemented ownership, Oracle reasoning, retrieval design, matched quality/cost, agent value, routing workload/error, population failures, volume-scale fallback economics and mandatory-review sensitivity. Historical E18 screenshots, mixed-population ladders, modeled reliability curves and E24 comparison plots remain preserved in their experiment directories but are not used as current headline evidence.

Reproduction. `python scripts/generate_reasoning_report_assets.py` regenerates the figures from frozen artifacts; `python scripts/build_reasoning_report.py` embeds every SVG and all CSS into the single final HTML; `python scripts/verify_reasoning_report.py` recomputes key metrics, validates links and confirms there are no external display dependencies. No paid inference is involved.

References

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